ELITE IGCSE ACADEMY

Student Past Paper Solutions

May 2025 4WM2H Worked Solutions

May2025 | 4WM2H | Unit 2

35 questions ▪ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION

Q#: 1 Marks: 3

TOPIC: **Statistics Toolkit**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

1 Rio has six cards.
He writes a number on each card so that

the range of the numbers is 10
the median of the numbers is 7.5
the mode of the numbers is 6

Rio arranges the cards so that the numbers are in order of size.

		6		10	14

Three of the numbers are hidden.
Complete the cards above to show the three numbers that are hidden.

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Statistics Toolkit**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Solution

The range is 10, so the first number is

$$14 - 10 = 4$$

For six numbers, the median is the mean of the third and fourth values.

$$\frac{6 + \text{fourth value}}{2} = 7.5$$

So the fourth value is 9. To make 6 the mode, the second value must also be 6.

FINAL ANSWER

the hidden numbers are $4, 6, 9$.

PAST PAPER QUESTION

Q#: 2 Marks: 3

TOPIC: **Volume & Surface Area**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

2 The diagram shows a solid prism with a cross section in the shape of a trapezium.

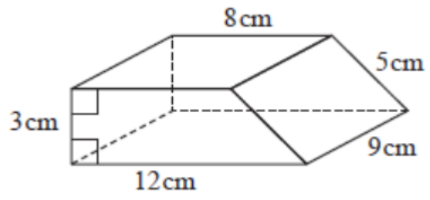

Diagram **NOT** accurately drawn

Work out the total surface area of the prism.

.... cm²

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Volume & Surface Area**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Solution

Area of one trapezium:

$$\frac{1}{2}(8+12)\times 3 = 30$$

Perimeter of the trapezium:

$$8+12+3+5=28$$

Rectangular faces:

$$28\times 9 = 252$$

Total surface area:

$$252+2(30)=312$$

FINAL ANSWER

312 cm^2.

PAST PAPER QUESTION

Q#: 3 Marks: 3

TOPIC: **Percentages**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

3 Bernice bought a car for \$8000
She sold the car for \$5600

Work out Bernice's percentage loss when she sold the car.

....%

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Percentages**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Solution

$$\text{loss} = 8000 - 5600 = 2400$$

$$\text{percentage loss} = \frac{2400}{8000} \times 100 = 30\%$$

FINAL ANSWER

30%.

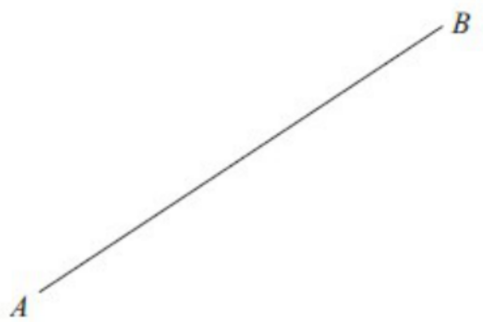
PAST PAPER QUESTION

Q#: 4 **Marks: 2**

TOPIC: **Bearings, Scale Drawing & Constructions**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

4 Using ruler and compasses only, construct the perpendicular bisector of the line *AB*
Show all your construction lines.



(Total for Question 4 is 2 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

Use a compass radius greater than half of AB. Draw matching arcs from A above and below the line, then draw matching arcs from B with the same radius. Join the two arc-intersection points.

FINAL ANSWER

the joined line is the perpendicular bisector of AB.

PAST PAPER QUESTION Q#: 5 Marks: 3

TOPIC: **Sequences**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

5 Here are the first five terms of a number sequence.

11 15 19 23 27

(a) Find an expression in terms of n for the nth term of this sequence.

....

(2)

The nth term of another number sequence is $5n + 4$

Salim says no term in this sequence is a prime number.

(b) Find a term in this sequence that shows Salim is wrong.

....

(1)

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 3

TOPIC: **Sequences**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Solution

The sequence increases by 4, so

$$u_n = 4n + 7$$

For the sequence $5n + 4$, take $n = 3$:

$$5(3) + 4 = 19$$

and 19 is prime.

FINAL ANSWER

$4n + 7$; a suitable counterexample is 19.

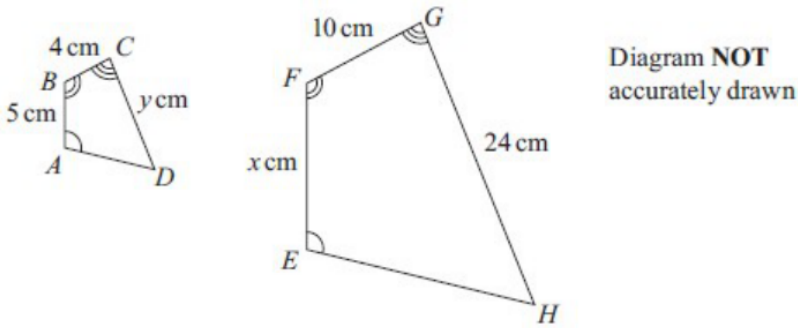
PAST PAPER QUESTION

Q#: 6 Marks: 4

TOPIC: **Congruence, Similarity & Geometrical Proof**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

6 The diagram shows two similar quadrilaterals, *ABCD* and *EFGH*



$AB = 5$ cm $BC = 4$ cm $CD = y$ cm

$EF = x$ cm $FG = 10$ cm $GH = 24$ cm

(a) Work out the value of x

$x =$ ....

(2)

(b) Work out the value of y

$y =$ ....

(2)

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 4

TOPIC: **Congruence, Similarity & Geometrical Proof**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Solution

The scale factor from $ABCD$ to $EFGH$ is

$$\frac{FG}{BC} = \frac{10}{4} = 2.5$$

So

$$x = EF = 5 \times 2.5 = 12.5$$

and

$$y = CD = \frac{24}{2.5} = 9.6$$

FINAL ANSWER

$x = 12.5$, $y = 9.6$.

PAST PAPER QUESTION

Q#: 7 Marks: 4

TOPIC: **Ratio Problem Solving**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

7 Pau, Sam and Tia share £240 in the ratios $3 : 4 : 5$

Sam and Tia each give £10 of their share to Pau.

Work out the ratios of the amounts of money that Pau, Sam and Tia now have.
Give your answer in its simplest form.

.... : .... : ....

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 4

TOPIC: **Ratio Problem Solving**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Topic check. Ratio Problem Solving. The tag is correct.

Method

The original ratio is

$$\text{Pau} : \text{Sam} : \text{Tia} = 3 : 4 : 5$$

There are

$$3 + 4 + 5 = 12 \text{ parts}$$

Each part is

$$240 \div 12 = 20$$

So the original amounts are

$$60, \ 80, \ 100$$

Sam gives 10 to Pau and Tia gives 10 to Pau, so the new amounts are

$$80, \ 70, \ 90$$

Divide by 10:

$$80 : 70 : 90 = 8 : 7 : 9$$

FINAL ANSWER

The new ratio is $8 : 7 : 9$.

PAST PAPER QUESTION

Q#: 8 Marks: 4

TOPIC: **Statistics Toolkit**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

8 The table gives information about the lengths of time, in minutes, that some people were in a coffee shop.

Length of time (m minutes)	**Frequency**
$0 < m \leqslant 10$	x
$10 < m \leqslant 20$	3
$20 < m \leqslant 30$	2
$30 < m \leqslant 40$	4
$40 < m \leqslant 50$	2

The information in the table is used to calculate an estimate for the mean length of time these people were in the coffee shop.

The estimate for the mean length of time is 20

Work out the value of x

x = ...

(Total for Question 8 is 4 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 4

TOPIC: **Statistics Toolkit**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Solution

Use class midpoints $5, 15, 25, 35, 45$.

$$\frac{5x + 15(3) + 25(2) + 35(4) + 45(2)}{x + 3 + 2 + 4 + 2} = 20$$

$$\frac{5x + 325}{x + 11} = 20$$

$$5x + 325 = 20x + 220$$

$$105 = 15x$$

$$x = 7$$

FINAL ANSWER

$x = 7$.

PAST PAPER QUESTION Q#: 9 Marks: 3

TOPIC: **Compound Interest & Depreciation**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

9 Tim buys a bracelet for 4000 Swiss francs.
The value of the bracelet increases by 7% each year.

Work out the value of the bracelet at the end of 3 years.
Give your answer correct to the nearest Swiss franc.

.... Swiss francs

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Compound Interest & Depreciation**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Topic check. Corrected to Compound Interest and Depreciation. This is repeated percentage growth for 3 years.

Method

The multiplier for a 7% increase is

$$1.07$$

After 3 years,

$$4000(1.07)^3 = 4900.172$$

Correct to the nearest Swiss franc,

$$4900.172 \approx 4900$$

FINAL ANSWER

4900 Swiss francs

PAST PAPER QUESTION Q#: 10 Marks: 3

TOPIC: **Simultaneous Equations**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

10 Solve the simultaneous equations

$$3x + 5y = 8$$
$$4x + y = -3.5$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Simultaneous Equations**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Solution

From

$$4x + y = -3.5$$

$$y = -3.5 - 4x$$

Substitute into $3x + 5y = 8$:

$$3x + 5(-3.5 - 4x) = 8$$

$$-17x = 25.5$$

$$x = -1.5$$

Then

$$y = -3.5 - 4(-1.5) = 2.5$$

FINAL ANSWER

$x = -1.5$, $y = 2.5$.

PAST PAPER QUESTION Q#: 11 Marks: 5

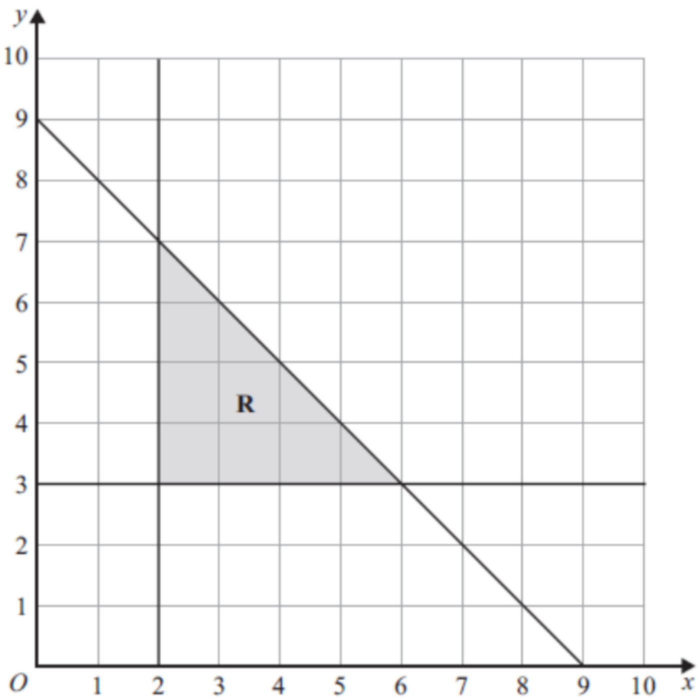
TOPIC: **Graphing Inequalities**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

11 (a) Solve the inequality $7 - 3t < 2t + 15$

....
(2)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down three inequalities that define the region **R**

....

....

....
(3)

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 5

TOPIC: **Graphing Inequalities**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Solution

(a)

$$7 - 3t < 2t + 15$$

$$-8 < 5t$$

$$t > -1.6$$

(b) The three boundary lines are $x = 2$, $y = 3$, and $x + y = 9$. The shaded side gives

$$x \geq 2, \qquad y \geq 3, \qquad x + y \leq 9$$

FINAL ANSWER

$t > -1.6$; $x \geq 2$, $y \geq 3$, $x + y \leq 9$.

PAST PAPER QUESTION

Q#: 12 **Marks: 3**

TOPIC: **Powers, Roots & Standard Form**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

12 (a) Write 8.6×10^{-5} as an ordinary number.

....

(1)

(b) Work out $(3\times10^{143}) \div (6\times10^{137})$

Give your answer in standard form.

....

(2)

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Solution

(a)

$$8.6 \times 10^{-5} = 0.000086$$

(b)

$$(3 \times 10^{43}) \div (6 \times 10^{37})$$

$$= 0.5 \times 10^{6} = 5 \times 10^{5}$$

FINAL ANSWER

0.000086, 5×10^{5}.

PAST PAPER QUESTION **Q#: 13** **Marks: 4**

TOPIC: **Circle Theorems**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

13 The diagram shows a circle, centre *O*

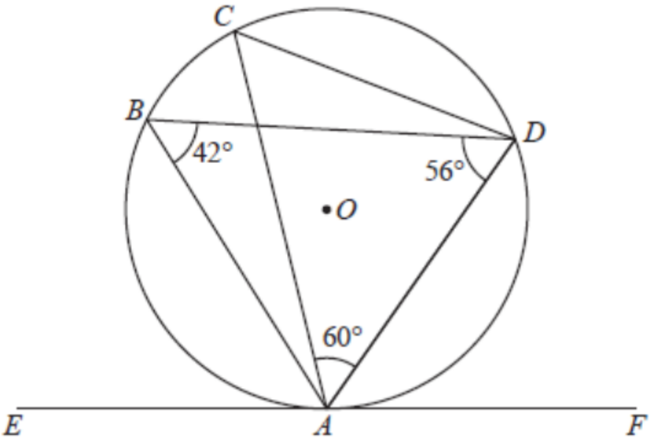

Diagram **NOT** accurately drawn

A, *B*, *C* and *D* are points on the circle.
EAF is the tangent to the circle at *A*

angle *ABD* = 42° **angle *BDA* = 56°** **angle *CAD* = 60°**

(a) (i) Find the size of angle *EAB*

....°

(1)

(ii) Give a reason for your answer to (a) (i)

....

....

(1)

(b) Work out the size of angle *CDB*

....°

(2)

(Total for Question 13 is 4 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 4

TOPIC: **Circle Theorems**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

By the alternate segment theorem,

$$\angle EAB = \angle ADB = 56^\circ$$

In triangle ABD,

$$\angle BAD = 180^\circ - 42^\circ - 56^\circ = 82^\circ$$

So

$$\angle BAC = 82^\circ - 60^\circ = 22^\circ$$

Angles subtended by the same chord CB are equal, so

$$\angle CDB = 22^\circ$$

FINAL ANSWER

56°, alternate segment theorem, 22°.

PAST PAPER QUESTION

Q#: 14 Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

14 The frequency table gives information about the times, in minutes, that 60 people took to complete a puzzle.

Time (t minutes)	**Frequency**
$0 < t \leqslant 10$	8
$10 < t \leqslant 20$	13
$20 < t \leqslant 30$	12
$30 < t \leqslant 40$	17
$40 < t \leqslant 50$	7
$50 < t \leqslant 60$	3

(a) Complete the cumulative frequency table.

Time (t minutes)	**Cumulative frequency**
$0 < t \leqslant 10$	
$10 < t \leqslant 20$	
$20 < t \leqslant 30$	
$30 < t \leqslant 40$	
$40 < t \leqslant 50$	
$50 < t \leqslant 60$	

(1)

(b) On the grid on the next page, draw a cumulative frequency graph for your table.

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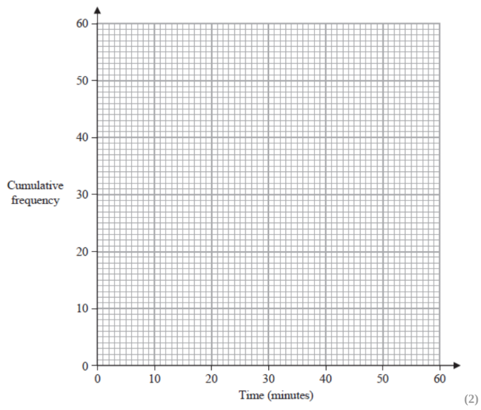

(2)

(c) Use your graph to find an estimate for the median time.

.... minutes

(1)

(d) Use your graph to find an estimate for the number of these people who took more than 45 minutes to complete the puzzle.

....

(2)

(Total for Question 14 is 6 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Solution

Cumulative frequencies:

$$8,\ 21,\ 33,\ 50,\ 57,\ 60$$

Plot the points

$$(10,8),(20,21),(30,33),(40,50),(50,57),(60,60)$$

The median is the 30th value. Interpolating in the 20 to 30 interval gives about 27.5 minutes. For more than 45 minutes, the estimate is about 7 people.

FINAL ANSWER

cumulative frequencies $8,21,33,50,57,60$; median about 27.5 minutes; more than 45 minutes about 7 people.

PAST PAPER QUESTION Q#: 15 Marks: 3

TOPIC: **Compound Interest & Depreciation**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

15 Fred buys a painting.

The value of the painting increases by 12% during the first year.
The value of the painting then decreases by 15% during the second year.

Work out the overall percentage decrease in the value of the painting during this two-year period.

....%

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Compound Interest & Depreciation**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

Use multipliers:

$$1.12 \times 0.85 = 0.952$$

So the final value is 95.2% of the original.

$$100 - 95.2 = 4.8$$

FINAL ANSWER

4.8% decrease.

PAST PAPER QUESTION Q#: 16 Marks: 2

TOPIC: **Statistics Toolkit**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

16 Here are the numbers of goals scored by a netball team in its 11 games this season.

7 8 10 11 13 14 16 17 19 21 24

Work out the interquartile range for the number of goals.

....

(Total for Question 16 is 2 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 2

TOPIC: **Statistics Toolkit**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

There are 11 values. Excluding the median 14:
Lower half: $7, 8, 10, 11, 13$, so $Q_1 = 10$.
Upper half: $16, 17, 19, 21, 24$, so $Q_3 = 19$.

$$\text{IQR} = 19 - 10 = 9$$

FINAL ANSWER

9.

PAST PAPER QUESTION Q#: 17 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

17 $A = 3^2 \times 5^4$ $B = 2^5 \times 3^2$ $C = 2^3 \times 5^6$

(a) Work out the value of $(ABC)^3$

Give your answer as a product of powers of its prime factors.

....

(2)

(b) Work out $(5 \times 10^{150}) \times (3 \times 10^{140})$

Give your answer as a product of powers of its prime factors.

....

(2)

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

(a)

$$ABC = (3^2 \times 5^4)(2^5 \times 3^2)(2^3 \times 5^6)$$

$$= 2^8 \times 3^4 \times 5^{10}$$

$$(ABC)^3 = 2^{24} \times 3^{12} \times 5^{30}$$

(b)

$$(5 \times 10^{150})(3 \times 10^{140}) = 15 \times 10^{290}$$

$$= 3 \times 5 \times (2 \times 5)^{290}$$

$$= 2^{290} \times 3 \times 5^{291}$$

FINAL ANSWER

$2^{24} \times 3^{12} \times 5^{30}$; $2^{290} \times 3 \times 5^{291}$.

PAST PAPER QUESTION

Q#: 18 Marks: 3

TOPIC: **Direct & Inverse Proportion**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

18 W is inversely proportional to y^2

$W = 50$ when $y = 4$

Find a formula for W in terms of y

....

(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 3

TOPIC: **Direct & Inverse Proportion**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

Since W is inversely proportional to y^2,

$$W = \frac{k}{y^2}$$

Using $W = 50$ when $y = 4$:

$$50 = \frac{k}{16}$$

$$k = 800$$

FINAL ANSWER

$W = \dfrac{800}{y^2}$.

PAST PAPER QUESTION **Q#: 19** **Marks: 4**

TOPIC: **Rearranging Formulas**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

19 Make t the subject of the formula $c = \dfrac{[illegible]}{7 - 8t^{2}}$

....

(Total for Question 19 is 4 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 4

TOPIC: **Rearranging Formulas**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

$$c = \frac{t^2 + 3}{7 - 8t^2}$$

$$c(7 - 8t^2) = t^2 + 3$$

$$7c - 3 = t^2(8c + 1)$$

$$t^2 = \frac{7c - 3}{8c + 1}$$

FINAL ANSWER

$t = \pm\sqrt{\dfrac{7c - 3}{8c + 1}}$.

PAST PAPER QUESTION

Q#: 20 Marks: 3

TOPIC: **Algebraic Proof**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

20 a, b and c are three consecutive even numbers where $a < b < c$

Prove algebraically that $b^2 = ac + 4$

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Algebraic Proof**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Solution

Let

$$a = 2n, \qquad b = 2n + 2, \qquad c = 2n + 4$$

Then

$$ac + 4 = (2n)(2n + 4) + 4$$

$$= 4n^2 + 8n + 4$$

$$= (2n + 2)^2 = b^2$$

So $b^2 = ac + 4$.

PAST PAPER QUESTION Q#: 20 Marks: 3

TOPIC: **Algebraic Proof**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

20 a, b and c are three consecutive even numbers where $a < b < c$

Prove algebraically that $b^2 = ac + 4$

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Algebraic Proof**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Solution
Let

$$a = 2n, \qquad b = 2n + 2, \qquad c = 2n + 4$$

Then

$$ac + 4 = (2n)(2n + 4) + 4$$

$$= 4n^2 + 8n + 4$$

$$= (2n + 2)^2 = b^2$$

So $b^2 = ac + 4$.

PAST PAPER QUESTION

Q#: 21 Marks: 7

TOPIC: **Functions**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

21 The function g is defined as

$$g(x) = \frac{7x + 20}{2x} \qquad x \neq 0$$

(a) Solve gg $(x) = 6$
Show clear algebraic working.

$x =$ ....
(3)

P81641A

The function h is defined as

$$h(x) = 5x^2 + 30x - 7 \qquad x \geqslant -3$$

(b) Express the inverse function h^{-1} in the form $h^{-1}(x) = \ldots$

$h^{-1}(x) =$ ....
(4)

(Total for Question 21 is 7 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 7

TOPIC: **Functions**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

(a) If $g(g(x)) = 6$, then

$$\frac{7g(x) + 20}{2g(x)} = 6$$

$$7g(x) + 20 = 12g(x)$$

$$g(x) = 4$$

So

$$\frac{7x + 20}{2x} = 4$$

$$7x + 20 = 8x$$

$$x = 20$$

(b)

$$h(x) = 5x^2 + 30x - 7 = 5(x + 3)^2 - 52$$

Let $y = 5(x + 3)^2 - 52$. Since $x \geq -3$, use the positive square root:

$$x + 3 = \sqrt{\frac{y + 52}{5}}$$

$$h^{-1}(x) = \sqrt{\frac{x + 52}{5}} - 3$$

FINAL ANSWER

$x = 20$; $h^{-1}(x) = \sqrt{\frac{x + 52}{5}} - 3$.

PAST PAPER QUESTION Q#: 21 Marks: 7

TOPIC: **Functions**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

21 The function g is defined as

$$g(x) = \frac{7x + 20}{2x} \qquad x \neq 0$$

(a) Solve gg$(x) = 6$
Show clear algebraic working.

x = ...
(3)

The function h is defined as

$$h(x) = 5x^2 + 30x - 7 \qquad x \geqslant -3$$

(b) Express the inverse function h^{-1} in the form $h^{-1}(x) = \ldots$

$h^{-1}(x)$ = ...
(4)

(Total for Question 21 is 7 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 7

Topic: **Functions**

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WORKED SOLUTION

Dr. Eslam Ahmed

Solution

(a) If $g(g(x)) = 6$, then

$$\frac{7g(x) + 20}{2g(x)} = 6$$

$$7g(x) + 20 = 12g(x)$$

$$g(x) = 4$$

So

$$\frac{7x + 20}{2x} = 4$$

$$7x + 20 = 8x$$

$$x = 20$$

(b)

$$h(x) = 5x^2 + 30x - 7 = 5(x + 3)^2 - 52$$

Let $y = 5(x + 3)^2 - 52$. Since $x \geq -3$, use the positive square root:

$$x + 3 = \sqrt{\frac{y + 52}{5}}$$

$$h^{-1}(x) = \sqrt{\frac{x + 52}{5}} - 3$$

FINAL ANSWER

$x = 20$; $h^{-1}(x) = \sqrt{\frac{x + 52}{5}} - 3$.

PAST PAPER QUESTION Q#: 22 Marks: 4

TOPIC: **Differentiation**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

22 Work out the range of values of x for which the graph of $y = x^3 + 2x^2 - 10x + 5$
has a positive gradient.
Show clear algebraic working.

....

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Differentiation**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

$$\frac{dy}{dx} = 3x^2 + 2x - 10$$

For a positive gradient,

$$3x^2 + 2x - 10 > 0$$

The roots are

$$x = \frac{-1 \pm \sqrt{31}}{3}$$

Since the quadratic opens upwards,

$$x < \frac{-1 - \sqrt{31}}{3} \quad \text{or} \quad x > \frac{-1 + \sqrt{31}}{3}$$

FINAL ANSWER

$x < \dfrac{-1 - \sqrt{31}}{3}$ or $x > \dfrac{-1 + \sqrt{31}}{3}$.

PAST PAPER QUESTION

Q#: 22 Marks: 4

Topic: **Differentiation**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

22 Work out the range of values of x for which the graph of $y = x^3 + 2x^2 - 10x + 5$ has a positive gradient.
Show clear algebraic working.

....

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Differentiation**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

$x < \dfrac{-1 - \sqrt{31}}{3}$ or $x > \dfrac{-1 + \sqrt{31}}{3}$.

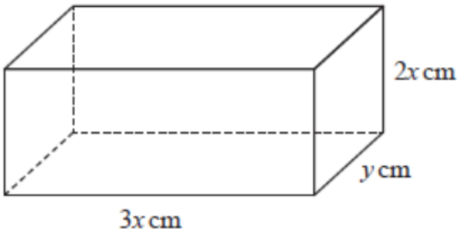
PAST PAPER QUESTION Q#: 23 Marks: 3

TOPIC: **Volume & Surface Area**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

23 The diagram shows a cuboid.

Diagram **NOT** accurately drawn



The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is 1014 cm^3
The total surface area of the cuboid is $A\text{ cm}^2$

Show that $A = 12x^2 + \dfrac{1690}{x}$

You must show all the stages of your working.

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 3

TOPIC: **Volume & Surface Area**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

$$3x \times 2x \times y = 1014$$

$$6x^2y = 1014$$

$$y = \frac{169}{x^2}$$

Surface area:

$$A = 2(3x \cdot 2x + 3x \cdot y + 2x \cdot y)$$

$$= 12x^2 + 10xy$$

$$= 12x^2 + 10x\left(\frac{169}{x^2}\right)$$

$$= 12x^2 + \frac{1690}{x}$$

FINAL ANSWER

$A = 12x^2 + \dfrac{1690}{x}$.

PAST PAPER QUESTION

Q#: 23 Marks: 3

TOPIC: **Volume & Surface Area**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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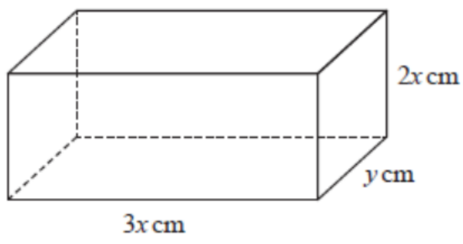

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Show that $A = 12x^2 + \dfrac{1690}{x}$

You must show all the stages of your working.

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 3

TOPIC: **Volume & Surface Area**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

$$3x \times 2x \times y = 1014$$

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$$= 12x^2 + 10xy$$

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$$= 12x^2 + \frac{1690}{x}$$

FINAL ANSWER

$A = 12x^2 + \dfrac{1690}{x}$.

PAST PAPER QUESTION Q#: 24 Marks: 4

TOPIC: **Sequences**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

24 The 21st term of an arithmetic series is 109
The sum of the first 52 terms of the series is 4381

Work out the 5th term of the series.
Show clear algebraic working.

....

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 4

TOPIC: **Sequences**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

For an arithmetic series,

$$a + 20d = 109$$

and

$$\frac{52}{2}(2a + 51d) = 4381$$

$$26(2a + 51d) = 4381$$

Solving gives

$$d = -4.5, \qquad a = 199$$

The fifth term is

$$a + 4d = 199 + 4(-4.5) = 181$$

FINAL ANSWER

181.

PAST PAPER QUESTION Q#: 24 Marks: 4

TOPIC: **Sequences**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

24 The 21st term of an arithmetic series is 109
The sum of the first 52 terms of the series is 4381

Work out the 5th term of the series.
Show clear algebraic working.

....

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 4

TOPIC: **Sequences**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

For an arithmetic series,

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The fifth term is

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FINAL ANSWER

181.

PAST PAPER QUESTION

Q#: 25 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

25 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm
The height of solid **B** is 18.6 cm

Given that

volume of solid **A** − volume of solid **B** = 735 cm^3

find the volume of solid **A**

.... cm^3

(Total for Question 25 is 4 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

Linear scale factor from solid A to solid B:

$$\frac{18.6}{31} = 0.6$$

So the volume scale factor is

$$0.6^3 = 0.216$$

Let the volume of A be V. Then

$$V - 0.216V = 735$$

$$0.784V = 735$$

$$V = 937.5$$

FINAL ANSWER

$937.5\ \text{cm}^3$.

PAST PAPER QUESTION

Q#: 25 **Marks: 4**

TOPIC: **Area & Volume of Similar Shapes**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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The height of solid **B** is 18.6 cm

Given that

volume of solid **A** − volume of solid **B** = 735 cm^3

find the volume of solid **A**

.... cm^3

(Total for Question 25 is 4 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

Linear scale factor from solid A to solid B:

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$$V = 937.5$$

FINAL ANSWER

$937.5\ \text{cm}^3$.

PAST PAPER QUESTION

Q#: 26 Marks: 5

Topic: **Vectors**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

26

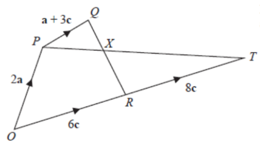


Diagram **NOT** accurately drawn

The diagram shows a quadrilateral *OPQR* in which

$\overrightarrow{OP} = 2\mathbf{a}$ $\overrightarrow{OR} = 6\mathbf{c}$ $\overrightarrow{PQ} = \mathbf{a} + 3\mathbf{c}$

(a) Find in terms of **a** and **c**

$\overrightarrow{QR} =$ ....

(1)

The point T is such that $\overrightarrow{RT} = 8\mathbf{c}$

The point X lies on QR such that PXT is a straight line.

(b) Using a vector method, find the ratio $QX : XR$

Give your answer in the form $m : n$ where m and n are integers.

Show your working clearly.

$QX : XR =$ ....

(4)

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 5

Topic: **Vectors**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

(a)

$$\overrightarrow{QR} = \overrightarrow{OR} - \overrightarrow{OQ}$$

$$\overrightarrow{OQ} = 2a + (a + 3c) = 3a + 3c$$

$$\overrightarrow{QR} = 6c - (3a + 3c) = 3c - 3a$$

(b) Let X divide QR so that

$$\overrightarrow{QX} = \lambda\overrightarrow{QR}$$

Then

$$\overrightarrow{OX} = 3a + 3c + \lambda(3c - 3a)$$

Also X lies on PT, where $\overrightarrow{OT} = 14c$, so

$$\overrightarrow{OX} = 2a + \mu(14c - 2a)$$

Equating coefficients gives $\lambda = \frac{5}{9}$. Therefore

$$QX : XR = \frac{5}{9} : \frac{4}{9} = 5 : 4$$

FINAL ANSWER

$\overrightarrow{QR} = 3c - 3a$; $QX : XR = 5 : 4$.

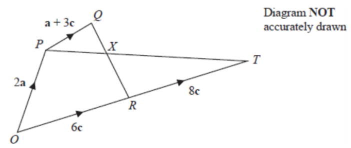
PAST PAPER QUESTION

Q#: 26 **Marks: 5**

TOPIC: **Vectors**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

26



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$\overrightarrow{QR}$ = ...
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Give your answer in the form *m* : *n* where *m* and *n* are integers.
Show your working clearly.

P81641A

QX :*XR* = ...
(4)

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 5

TOPIC: **Vectors**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

(a)

$$\overrightarrow{QR} = \overrightarrow{OR} - \overrightarrow{OQ}$$

$$\overrightarrow{OQ} = 2a + (a + 3c) = 3a + 3c$$

$$\overrightarrow{QR} = 6c - (3a + 3c) = 3c - 3a$$

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$$QX : XR = \frac{5}{9} : \frac{4}{9} = 5 : 4$$

FINAL ANSWER

$\overrightarrow{QR} = 3c - 3a$; $QX : XR = 5 : 4$.

PAST PAPER QUESTION

Q#: 27 **Marks: 4**

TOPIC: **Transformations of Graphs**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

27 A curve has equation $y = f(x)$

There is one maximum point on this curve.
The coordinates of this maximum point are $(6, -5)$

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x - 4)$

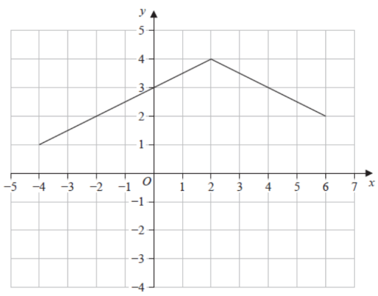
(.............................. , ...)
(1)

(ii) $y = f(3x)$

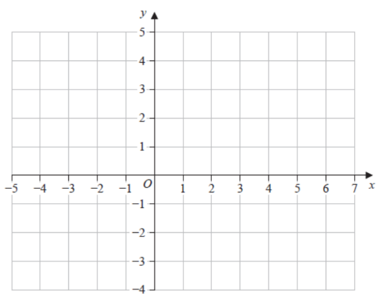
(.............................. , ...)
(1)

P81641A

Here is the graph of $y = g(x)$



(b) On the grid below, draw the graph of $y = g(2x) - 3$



(2)

(Total for Question 27 is 4 marks)

PAST PAPER SOLUTION

Q#: 27 Marks: 4

TOPIC: **Transformations of Graphs**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Solution

(a)

For $y = f(x-4)$, move the graph 4 units right:

$$(6, -5) \to (10, -5)$$

For $y = f(3x)$, divide the x-coordinate by 3:

$$(6, -5) \to (2, -5)$$

(b) For $y = g(2x) - 3$, halve each x-coordinate and subtract 3 from each y-coordinate. Useful transformed points are

$$(-4, 1) \to (-2, -2), \quad (2, 4) \to (1, 1), \quad (6, 2) \to (3, -1)$$

Join these points with straight line segments.

FINAL ANSWER

$(10, -5)$, $(2, -5)$, graph through $(-2, -2), (1, 1), (3, -1)$.

PAST PAPER QUESTION

Q#: 27 **Marks: 4**

TOPIC: **Transformations of Graphs**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

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P81641A

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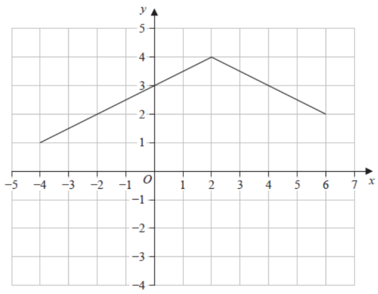

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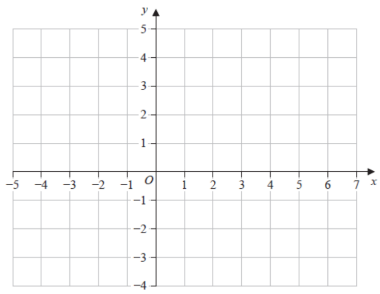

(2)

(Total for Question 27 is 4 marks)

PAST PAPER SOLUTION

Q#: 27 Marks: 4

TOPIC: **Transformations of Graphs**

May 2025 4WM2H - May2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

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$$(-4,1) \to (-2,-2), \quad (2,4) \to (1,1), \quad (6,2) \to (3,-1)$$

Join these points with straight line segments.

FINAL ANSWER

$(10,-5)$, $(2,-5)$, graph through $(-2,-2),(1,1),(3,-1)$.